

Atomistic Insights into Lithium–Glyme Solvate Ionic Liquids: Effects of Chain Length and Anion Coordination

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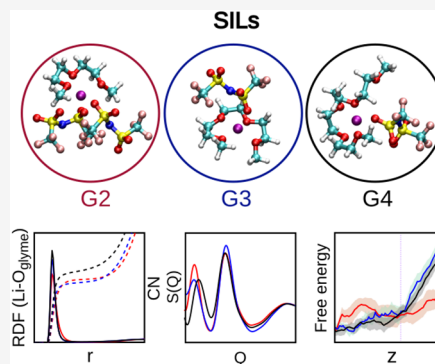


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ABSTRACT: Mixtures of lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) in diglyme (G2), triglyme (G3) and tetraglyme (G4) at solvate ionic liquid (SIL) concentrations were investigated using classical molecular dynamics (cMD) simulations with a physically motivated force-field specifically developed for modeling these systems. The structural and dynamical properties of the mixtures were computed and analyzed. Lithium solvation shells, radial distribution functions, and X-ray structure factors were studied across the different SIL systems. Translational diffusion and rotational relaxation times were also evaluated, exhibiting similar trends with increasing glyme chain length. The results are consistent with experimental data and in good agreement with previous computational studies on G3 and G4. These findings validate the accuracy of the force field in modeling glyme systems and its use for describing the $[\text{Li}(\text{G2})_{4/3}][\text{TFSI}]$ mixture. Additionally, the thermal and electrochemical stability of these electrolytes were systematically examined. The thermal stability appears to be governed by cooperative interactions among glyme molecules, while the electrochemical stability is primarily influenced by Li^+ -anion interactions, which vary significantly with glyme chain length. Overall, the study sheds light on the crucial role of the anion in these glyme-based SILs and offers valuable insights into Li^+ -glyme systems at SIL concentrations, highlighting their promise as potential Li-ion battery electrolytes.



INTRODUCTION

Solvated ionic liquids (SILs) typically consist of a lithium salt, such as lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), dissolved in oligo(ethylene glycol) dimethyl ethers such as triglyme (G3) or tetraglyme (G4) at an equimolar concentration.¹ Glyme molecules coordinate with Li^+ to form a ‘supercation’ due to the chelating effect of the solvent, which results in a mixture with an overall behavior resembling traditional room temperature ionic liquids.^{2,3} SILs have been proposed as novel electrolytes for electrochemical devices due to their high thermal and electrochemical stability, low flammability and toxicity.^{1,4–7} However, their practical implementation faces significant challenges, including low conductivity caused by high viscosity and the increased production costs associated with using larger amounts of lithium compared to traditional electrolytes.^{8,9} Therefore, understanding the relationship between the structure and dynamics of these glyme-based electrolytes is crucial in order to further develop them for their use in lithium-ion batteries (LIBs). However, there are relatively few studies on SILs due to the complexity in the molecular characterization of the Li^+ -glyme systems, thereby making experimental investigations challenging.¹⁰ Hence, most studies in this area have used molecular dynamics (MD) simulations to investigate molecular level details.^{11,12} For example, lifetime measurements of Li-glyme and ion pairs have been calculated using both nonpolarizable¹¹ and polarizable force fields¹³ for the G4

system. Interestingly, the residence times obtained with the two methodologies differ by around one order of magnitude, even though both MD approaches successfully reproduce the experimental trends in self-diffusivities. Consequently, new studies are needed to further explore the ionic speciation of the system as a function of the glyme length. Additionally, since the latter can be used to tune electrolyte properties,¹⁴ investigating new glyme-based electrolytes could provide valuable insights into the interplay between glyme and anion within the lithium solvation shell.

SILs of LiTFSI and G3 or G4 have been studied both experimentally and theoretically via molecular simulations,^{2–8,15–18} while electrolytes consisting of LiTFSI in diglyme (G2), particularly at SIL concentrations, have received relatively little attention. The primary reason is the complex between Li^+ and G2 cannot be formed in a stoichiometric ratio of one to one, which is one of the defining criteria of a SIL.¹⁹ However, some studies have investigated G2 at high salt concentrations and found that LiTFSI and G2 may form SIL-

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like liquids. For instance, Zhang et al. have proposed that the $[\text{Li}(\text{G2})_{4/3}][\text{TFSI}]$ molten mixture (around 0.43 molar fraction) could be regarded as a SIL.²⁰ In contrast, Choquette et al. studied LiTFSI solvated in different glymes and different concentrations, finding that G2 forms a complex with LiTFSI for molar fractions below 0.4, but becomes amorphous above that value.²¹ Other experimental works have characterized the saturated mixture of LiTFSI and G2, revealing strong Coulombic interactions between solvated ions, minimal aggregation, and the absence of free solvent molecules.²² Computational studies are, on the other hand, very limited. Arai et al. investigated the Li^+ conduction in SILs for different glyme chain lengths, considering equimolar mixtures between glymes and LiTFSI.²³ To the best of our knowledge, no computational studies have been conducted on the specific $[\text{Li}(\text{G2})_{4/3}][\text{TFSI}]$ electrolyte.

In this work, classical MD (cMD) simulations were carried out on the three SILs composed of LiTFSI solvated in G2, G3, and G4 solvents, utilizing a recently developed force field, which is specifically tailored for modeling the interactions between LiTFSI and glymes.¹⁰ The aim of this study is to computationally characterize these mixtures and benchmark key molecular properties, and shed some light about the molecular aspects that give rise to their dynamical, thermal and electrochemical behavior. This paper is organized as follows: it starts with a section detailing the computational methods, followed by an Analysis and Results section that presents the structural and dynamical properties derived from the simulations. Next, it includes a free-energy study of the electrolyte-air interphase for each mixture and presents the results from the highest occupied molecular orbital–lowest unoccupied molecular orbital (HOMO–LUMO) energy gap analysis. Finally, conclusions with a summary of the relevant findings are presented at the end of the article.

■ COMPUTATIONAL METHODOLOGY

Molecular Dynamics Simulations. Classical molecular dynamics (cMD) simulations of LiTFSI solvated in G2, G3, and G4 at SIL concentration were performed using a modified PCFF force field previously developed by Carrillo-Bohórquez et al.¹⁰ The PCFF force-field is a standard class2 effective two-body force-field. The modified force field was parametrized to reproduce salt-glyme interactions by fitting forces and torques on the ions from *ab initio* MD simulations of LiTFSI in diglyme. The resulting model accurately captures *ab initio* structural features, experimental diffusion trends, and X-ray structure factors. It also exhibits good transferability across different glymes and salt concentrations as previously demonstrated.¹⁰ All simulations were carried out using the LAMMPS software package.²⁴ A time step of 1 fs was used along with the velocity-Verlet integration algorithm. The Nose-Hoover thermostat was applied for both the canonical (NVT) and isothermal–isobaric (NPT) ensembles, with a temperature damping parameter of 0.1 ps. For the NPT simulations, an isotropic Nose-Hoover barostat was implemented with a pressure of 1 atm and a damping parameter of 1 ps. A cutoff distance of 6 Å was used for real-space Coulombic and Lennard-Jones interactions, and the particle–particle–particle-mesh (PPPM) method was used for long-range electrostatic interactions, with a relative force error set to 10^{-6} . Periodic boundary conditions were applied in all directions.

Simulation Setup and Equilibration. Cubic simulation boxes with side lengths of around 40 Å for the various glymes

were constructed using the Packmol package²⁵ and glyme SIL structures to accelerate the convergence of the calculations.¹⁰ For G2, the $[\text{Li}(\text{G2})_{4/3}][\text{TFSI}]$ mixture ratio reported by Zhang et al. was adopted,²⁰ and initial structures were generated with either one or two glymes coordinated to a single Li^+ . In these structures, approximately half of the Li^+ were coordinated to a single glyme, while the remaining Li^+ were coordinated to two glymes. Equilibrations can present a significant challenge for viscous systems such as SILs and hence a charge scaling procedure was adopted. Equilibration was performed in the NPT ensemble at 300 K, with ion charges being systematically scaled as follows. The equilibration process, carried out in the NPT ensemble, began with an energy minimization and a 6 ns run using a scaling factor of 0.5 for the charges, followed by a similar run with a scaling factor of 0.6. This procedure was repeated with scaling factors of 0.7, 0.8, 0.9, and 1.0, resulting in a total equilibration time of 36 ns. The production stage consisted of a 400 ns run in the NVE ensemble. For the G3 and G4 simulations, the usual $[\text{Li}(\text{G3})_1][\text{TFSI}]$ and $[\text{Li}(\text{G4})_1][\text{TFSI}]$ ratios were used, respectively, with SIL structures serving as initial configurations, as mentioned earlier.¹⁰ The equilibration process for these glyme systems involved an energy minimization and an annealing procedure that heated the system from 1 to 400 K over 2 ns in the NVT ensemble, followed by a 10 ns run at constant temperature (400 K) and 1 atm in the NPT ensemble. The system was then cooled from 400 to 300 K over 2 ns in the NVT ensemble, concluding with a final stage in the NPT ensemble at 300 K and 1 atm for 10 ns. Finally, simulations of 400 ns in the NVE ensemble were run and used as production. Details on the number of molecules and final box sizes for all glymes are available in Table S1 of the Supporting Information (SI). The production runs were then used in subsequent analyses.

Radial distribution functions (RDFs) were obtained with the VMD software.²⁶ The positions of the first minima in the RDFs for $\text{Li}-\text{O}_{\text{glyme}}$ and $\text{Li}-\text{O}_{\text{TFSI}}$ were used to define the coordination of Li^+ to glyme and TFSI[−] species, respectively, within the first solvation shell. The distributions of glyme and anion molecules coordinated to Li^+ , along with the conformational maps of glyme oxygen angles versus tail-to-tail distances, were computed using in-house Fortran codes.

Diffusion coefficients were computed through the TRAVIS package,^{27,28} according to the formula

$$D = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d}{dt} \text{MSD}(t) = \frac{1}{6} \lim_{t \rightarrow \infty} \frac{d}{dt} \left\langle \sum_{i=1}^N [\mathbf{r}_i(t) - \mathbf{r}_i(0)]^2 \right\rangle \quad (1)$$

where $\text{MSD}(t)$ is the mean squared displacement, N is the total number of atoms or molecules of a given species, and $\mathbf{r}_i(t)$ and $\mathbf{r}_i(0)$ are the position vectors of the i -th atom or molecule at a time t and at the reference origin time ($t = 0$), respectively. The angular brackets in eq 1 indicates an ensemble average. In the case of the glyme and anion molecules, the center of mass positions were used to calculate the MSD. The maximum lag time considered was 40 ns.

Rotational dynamics were investigated for both the anion and glyme species. For the anion, the orientation of TFSI[−] was defined using a unit vector directed along the sulfur–sulfur axis, while for the glyme molecules, a unit vector was aligned with the direction of the carbon atoms in the methyl groups (Figure 1). The time correlation function describing the

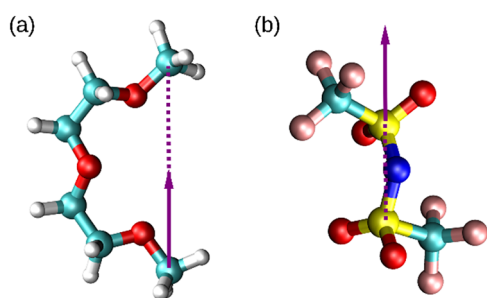


Figure 1. Unit vectors defined for (a) glyme and (b) anion molecules, used to analyze their rotational dynamics. Carbon, oxygen, hydrogen, sulfur, nitrogen and fluorine atoms are colored cyan, red, white, yellow, blue, and pink, respectively.

reorientation of each unit vector was calculated by determining^{29,30}

$$r(t) = \langle P_2[\cos \theta(t)] \rangle \quad (2)$$

where $P_2[\cos \theta(t)]$, with $\cos \theta(t) = \hat{\mathbf{u}}(t) \cdot \hat{\mathbf{u}}(0)$, is a Legendre polynomial of order 2, $\hat{\mathbf{u}}(t)$ and $\hat{\mathbf{u}}(0)$ represent the unit vector $\hat{\mathbf{u}}$ at time t and time zero, respectively, and the angular brackets represent an ensemble average. Equation 2 provides an approximation based on the assumption of an isotropic molecule, which serves as a useful reference for comparing simulated results with NMR relaxation³¹ and with reorientation times obtained from polarized infrared spectroscopy^{29,30}. The resulting function $r(t)$ was then fit to a triexponential function and integrated over time to yield τ_r , the rotational relaxation time³¹

$$\tau_r = \int_0^\infty r(t) dt = \int_0^\infty \left(\sum_{i=1}^3 a_i e^{-t/b_i} \right) dt \quad (3)$$

In eq 3, a_i and b_i are fitting parameters, and the resultant integral can be evaluated analytically.

The X-ray structure factors were calculated using the TRAVIS software, based on the final 50 ns of the production runs. A maximum radial distribution function distance of 21 Å and a wave vector modulus limit of 200 nm⁻¹ were applied. These calculations were based on the following expression³²

$$S(Q) = \frac{\rho_0 \sum_i \sum_j x_i x_j f_i(Q) f_j(Q) \int_0^\infty 4\pi r^2 (g_{ij}(r) - 1) \frac{\sin(Qr)}{Qr} dr}{[\sum_i x_i f_i(Q)]^2} \quad (4)$$

Here, ρ_0 represents the bulk density, with i and j denoting atom types, x and f referring to atomic fractions and atomic form factors, respectively, and $g_{ij}(r)$ representing the partial radial distribution function between atoms i and j .³²

Umbrella Sampling. The umbrella sampling technique was used to calculate the free energy profile of a glyme across the electrolyte-air interface for each system. To simulate this interface, simulation boxes of size approximately (40 × 40 × 100) Å³ were constructed (Figure 2), containing an approximately cubic electrolyte of about 40 Å in length in contact with an air interface. These cubic structures, taken from the cMD simulations described above, served as the initial configurations for the electrolytes. The number of molecules for each glyme system can be found in Table S2 of the SI. Periodic boundary conditions were applied in all

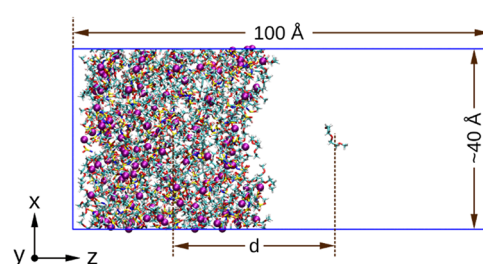


Figure 2. Rectangular simulation box used in umbrella sampling calculations, where d represents the distance between the center of mass of the bulk region and the center of mass of the displaced glyme. Atom coloring is the same as in Figure 1, with lithium ions in purple.

directions. For each system, a glyme molecule was moved from the interior of the electrolyte to the air region. The process began with a rapid scan in the NVT ensemble, displacing the glyme in steps (or windows) of 1 Å, allowing for 1 ns of equilibration per window at 300 K. The distance, in the z -direction, between the center of mass of the electrolyte and that of the glyme molecule was used as the collective variable, ranging from 0 to 33 Å. Each window was then further equilibrated for 10 ns, during which the distance d between the center of mass of the displaced glyme and that of the bulk electrolyte in the z -direction (as shown in Figure 2), under the influence of a harmonic bias potential, was recorded. In the case of G3, the system was equilibrated for 20 ns. After equilibration, a production of 10 ns was run per window in each glyme system. The COLVARS package³³ implemented in LAMMPS was used to carry out these simulations, applying a harmonic bias potential with a spring constant of 10 kcal/mol/Å² for each window. Positions were recorded every 0.1 ps. Overlap between the density distributions was verified through inspection and are shown in Figure S1 of the SI. Finally, the free energy profiles were generated from the position distributions using the weighted histogram analysis method (WHAM) algorithm³⁴ with 500 bins and a convergence tolerance of 10⁻⁸ kcal/mol. The details of the equilibration and production stages are provided in the SI (Figure S2). The standard deviations of the free energy profiles were calculated by dividing the data into four equal blocks of 2.5 ns each and applying the block averaging method.

Static Electronic Structure Calculations. The gap energies between the HOMO and LUMO levels were calculated using density functional theory (DFT) single-point calculations on the most representative solvation structures from each electrolyte. These structures were directly extracted from the production stages of the previously described MD simulations. A total of 15 structures per major solvation shell were randomly selected from the trajectory files. The calculations, without any prior geometry optimization, were performed using the Gaussian 16 software package,³⁵ with the PBE functional³⁶ and the 6-31+G(d,p) basis set.³⁷ This level of theory has been shown to provide results with accuracy comparable to the MP2 method.³⁸ Furthermore, a dielectric constant of ~16 was incorporated using the conductor-like polarizable continuum model (CPCM)^{39,40} to simulate an ionic liquid environment.⁴¹ The inclusion of a dielectric medium is essential for accurately calculating LUMO energies, as has been highlighted before.⁴² As result, a total of 15 single-point calculations were performed for each major structure, and the HOMO–LUMO gap energies were obtained as the average of the 15 gap values for each structure.

RESULTS AND DISCUSSION

Solvation Shell Structure. The structures of the different LiTFSI-glyme electrolytes were first investigated using radial distribution functions (RDFs). RDFs for different moieties of the LiTFSI and glyme (G2, G3, and G4) are illustrated in Figure 3. For the Li–O_{glyme} RDF, the first peak occurs close to

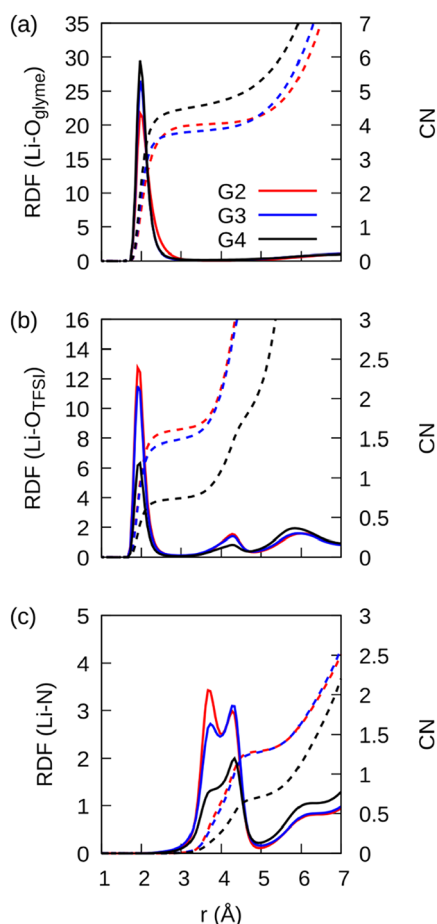


Figure 3. Radial distribution function and coordination numbers, RDFs, for (a) Li–O_{glyme}, (b) Li–O_{TFSI} and (c) Li–N interactions, for G2 (red), G3 (blue) and G4 (black) solvents. Coordination numbers (CN) are presented in dashed lines.

2.0 Å in all the electrolytes under study (Figure 3a), with the peak for G4 being the highest, followed by G3 and G2. The first minima of the Li–O_{glyme} RDF are observed at 3.8, 3.6, and 3.5 Å for G2, G3, and G4, respectively. Notably, similar coordination numbers (CN) of glyme oxygen atoms around the Li⁺ are observed for G2 and G3, but their underlying solvation structures are different. For G2, glyme molecules coordinate Li⁺ with either one or two molecules, resulting in an overall CN of 4.0, while G3 coordinates Li⁺ with its four oxygen atoms, leading to a higher and narrower first peak when compared with G2. These results demonstrate the smaller and more structured first solvation shell of G3 when compared to G2, but with a similar CN of oxygen atoms of 3.9. In comparison, G4 coordinates Li⁺ with either four or five oxygen atoms, resulting in a CN of 4.6. The RDFs for Li–O_{TFSI} (Figure 3b) show a first peak position at around 2.0 Å for all the SILs, with peak heights following the opposite trend to that observed for the Li–O_{glyme} RDFs, and minima around 3.0 Å in all systems. Furthermore, the CNs of the Li–O_{TFSI} interaction

are 1.6, 1.5, and 0.8 for G2, G3 and G4, respectively, indicating that the anion completes the expected Li–O CN of around 6 through a mono- or bidentate coordination that depends on the glyme length.⁴³ This observation is further supported by examining the RDFs and CNs for the Li–N atom pair. In these RDFs (Figure 3c), it is observed that the Li–N interaction features a profile with two peaks before reaching the first minimum at around 5.0 Å in all the electrolytes. The RDF peak maxima appear around 3.7 and 4.3 Å in the electrolytes. While the first peak corresponds to bidentate coordination of the anion and decreases in height as glyme length increases, the second peak reflects monodentate coordination, with the height being the lowest for G2 and similar for G3 and G4. This double-peak feature in the RDFs of Li–N provides some information about the orientation of TFSI[−], which is primarily directed toward the Li⁺ in G2 and becomes more outward-oriented as the glyme length increases. Notably, the CNs are similar for all glymes: 1.3 for G2 and G3, and 1.2 for G4. It is important to note that the RDFs for G2 and G3 in this study show lower and slightly shifted peaks compared to OPLS-based simulations (Figure S3 of the SI).⁴⁴

A partial validation of the RDFs was assessed by comparing simulated X-ray scattering patterns of the Li⁺–glyme systems with experimental results. The computational structure factors are shown in Figure 4, alongside reported data from small-angle X-ray scattering (SAXS) and high-energy X-ray diffraction (HEXRD) experiments for G2,²² and for G3 and G4,² respectively.

Figure 4 shows that the characteristic double-peak profile of the TFSI-based ILs² is present in all the SIL electrolytes studied here. The first peak at ~ 1.5 Å^{−1} corresponds to intermolecular distances and it is also observed in the pure solvents, while the second peak around 0.9 Å^{−1}, which corresponds to a distance of ~ 7 Å in real space, is associated with the separation between the Li⁺/glyme complex and the anion.⁴⁵ This peak increases as the salt concentration rises.^{45,46} In the case of G2, the structure factor from cMD presents peaks at 0.9 and 1.6 Å^{−1}, whereas the SAXS experiment peaks are found at 0.9 and 1.4 Å^{−1}, respectively (Figure 4a). Note that the comparison between the experimental and theoretical profiles in the case of G2 is rather qualitative, as they correspond to different concentrations, and hence are not further discussed.²² In the case of G3 and G4, the 1:1 molar ratio used in the HEXRD experiments may not accurately represent the chemical composition in the molten state but instead reflects the intended formula for the crystalline complexes.² For G3, the cMD profile displays peaks at 0.9 and 1.5 Å^{−1}, while the HEXRD shows peaks close to the same positions (Figure 4b). For G4, the cMD peaks are at 1.0 and 1.5 Å^{−1}, compared to 0.9 and 1.5 Å^{−1} in the experimental data (Figure 4c). Overall, the simulations capture semiquantitatively the key scattering features, with the positions of the first peaks closely matching the reported values. This agreement suggests that the simulations accurately reproduce the structure size in the middle-Q region peak (0.9–1.0 Å^{−1}), which arise from long-range interactions due to the addition of LiTFSI to the glymes.^{5,45,46} While there are some more noticeable discrepancies in the peak positions within the high-Q region ($Q > 1.0$ Å^{−1}) associated with intra and interatomic distances in liquids,⁵ the general shape of these peaks along with the position of their local minimum around 2.1 Å^{−1} agree well with the experimental data (Figure 4b,c). The peak positions in the G3 SIL are smaller than those of the G4 SIL, suggesting the

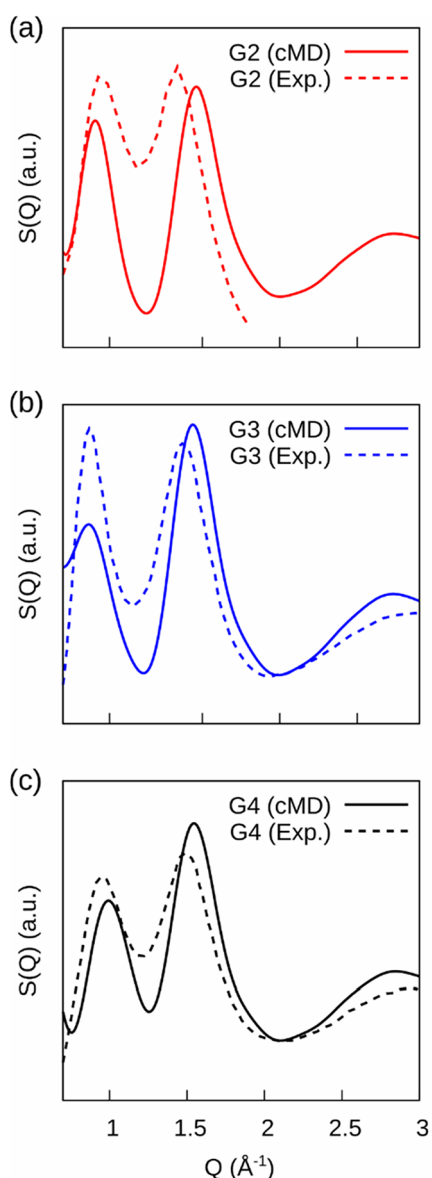


Figure 4. Solid lines: Simulated X-ray structure factors from classical molecular dynamics (cMD) calculations for (a) $[\text{Li}(\text{G2})_{4/3}][\text{TFSI}]$, (b) $[\text{Li}(\text{G3})_1][\text{TFSI}]$, and (c) $[\text{Li}(\text{G4})_1][\text{TFSI}]$. Dashed lines: Experimental data from SAXS for a saturated LiTFSI concentration in G2²² (a), and HEXRD data obtained at 298 K for the $[\text{Li}(\text{G3})_1][\text{TFSI}]$ (b) and $[\text{Li}(\text{G4})_1][\text{TFSI}]$ (c) electrolytes.²

presence of larger Li-solvated clusters in G3 compared to G4.² This observation is consistent with the findings of the Li^+ solvation shell, where one or more anions are more prevalent in G3 than in G4. Additionally, peak intensities have been correlated with the relative contact ion-pair (CIP) content of the electrolytes since higher peaks indicate greater CIP formation.² Consequently, it can be concluded that G3 has a greater CIP content than G4, which is consistent with the analysis of the solvation shells as described below. Finally, the individual components of the calculated X-ray structure factor, along with the partial structure factors, are provided in Figure S4 of the SI.

The solvation shell structure and speciation of Li^+ is better described by the distributions of coordinated glyme molecules and TFSI^- per lithium center (Figure 5). Analysis of these distributions reveals that the predominant solvation structures

in G2 have either one or two glymes coordinating a single Li^+ . Specifically, it was found that 57% of the Li^+ coordinate with a single glyme molecule, while the rest is mostly coordinated with a pair of glyme molecules (Figure 5). Furthermore, approximately 38% of the Li^+ are not coordinated with anions in G2, given that two glyme molecules participate in their solvation structure (Figure 5), providing a total of six coordinating oxygen atoms. In contrast, a single G4 molecule, with its five oxygen atoms, can fully coordinate a Li^+ ion on its own (Figure 5). For G3, which has only four oxygen atoms, an anion must contribute to achieve the typical Li–O coordination number of approximately 5–6.¹¹ This requirement leads to the predominance of solvation structures involving one glyme molecule and one anion and explains the low percentage of fully glyme-solvated cations observed in the G3 electrolyte (Figure 5). Note that the results of this work are in agreement with those of Harte et al.⁴⁴ Therefore, a difference arises in the coordination of TFSI^- oxygen atoms to Li^+ in G3, where more than 94% of the Li^+ are coordinated with TFSI^- , whereas the value decreases to $\sim 61\%$ for G4. The two major solvation structures for G3 correspond to Li^+ coordination to one G3 and one TFSI^- (70.6%), and one G3 and two TFSI^- (17.4%). In the case of G4, the major structures correspond to Li^+ coordination to a single G4 (34.2%), and one G4 and one TFSI^- (56.2%). These results are again consistent with those previously reported by Harte et al., although the fraction of Li^+ coordinated to five G4 oxygen atoms found in that work is higher.⁴⁴ This difference could arise from the overcoordination of the OPLSA-based force field,⁴⁴ as seen in previous work.⁴⁷ Major Li^+ solvation structures that depict the cases of highest probability are shown in Figure 5d. The high percentage of negatively charged solvation structures in these systems (two or more anions in the solvation shell of Li^+) is attributed to a significant fraction of anions coordinating with two or more Li^+ ions, which bridge across different solvation shells and form ionic networks (Table S3 in the SI). Unlike Li^+ , the anion contains five spatially separated coordination sites (four oxygen atoms and one nitrogen atom), allowing it to bridge multiple Li^+ solvation structures. As shown in Table S3, each anion typically coordinates with one or two Li^+ ions. However, in the G4 system, the size of the solvent molecule and weaker cation–anion interaction limit the coordination of the anion, resulting in anions being associated with a single Li^+ . This behavior contributes to a higher fraction of unbound anions and a greater percentage of fully glyme-solvated cations in G4 SIL. In contrast, in G2 and G3, the lower number of oxygen atoms per glyme molecule leads to a larger participation of the anion in Li^+ solvation shell and consequently resulting in fewer unbound anions when compared to the G4 (Table S3). Additionally, it is observed that a low percentage of free glymes was present in each system: 2.8% of G2, 0.6% of G3, and 1.4% of G4. The low amount of uncoordinated solvent molecules satisfies the key criteria for the classification of these systems as SILs.¹⁹

Further insight into the Li^+ solvent coordination was gained by analyzing the conformation of coordinated glymes, using the angles between oxygen atoms a_i , with $i = 1, 2, 3, 4$ (Figure 6a) and tail-to-tail distances r_i —defined by the separation between the terminal methyl carbon atoms. Note that G2 and G3 contain one type of internal angle, but G4 requires an additional inner angle, a_4 , to completely characterize its backbone conformation (Figure 6a). Figure 6b–e display the

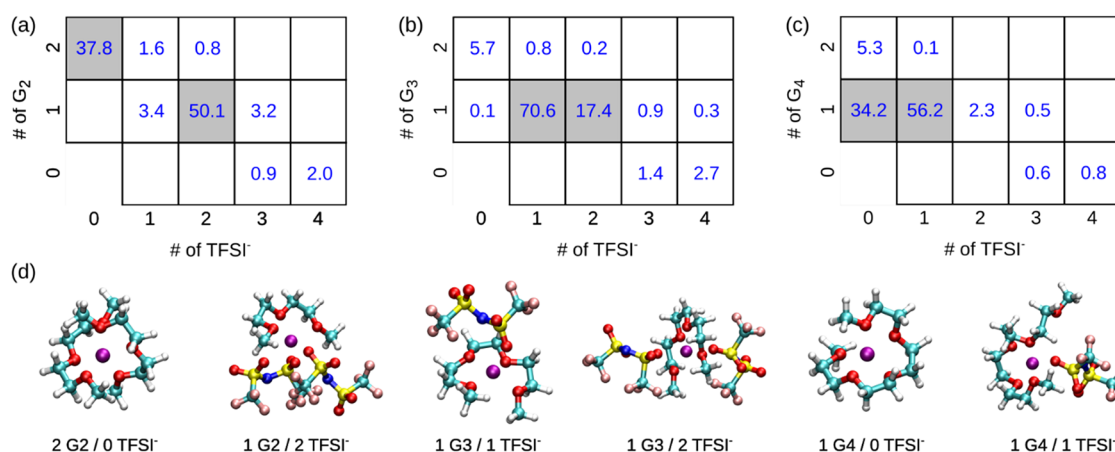


Figure 5. (a–c) Joint normalized probability (in percentage) of coordinated Li^+ as a function of the number of glymes molecules and TFSI^- in the first solvation shell for the (a) $[\text{Li}(\text{G}_2)_{4/3}][\text{TFSI}]$, (b) $[\text{Li}(\text{G}_3)_1][\text{TFSI}]$, and (c) $[\text{Li}(\text{G}_4)_1][\text{TFSI}]$ mixtures. The two highest percentages, corresponding to the major structures in each system, are highlighted with a gray background. (d) Major solvation structures found in the different glyme electrolytes.

joint normalized probability distributions obtained for each r_t vs a_i across their corresponding glyme. The G_2 electrolyte (Figure 6b) presents a single distribution with a prominent and sharp peak at around $r_t = 6.5$ Å and $a_1 = 110^\circ$ indicating a complete coordination of the three oxygen atoms of G_2 to a Li^+ (structure I of Figure 6f). This G_2 conformation is due to being primarily in the Li^+ solvation shell (Figure 5d). In the case of G_3 (Figure 6c), the conformational probability features a more complex distribution with one main peak around $r_t = 4.0$ Å and $a_2 = 110^\circ$, along with a small and broad distribution appearing between 7 and 11 Å in r_t , and as the tail of the main peak in a_2 . While the sharp peak represents a fully coordinated and dominant structure of G_3 glyme with Li^+ (structure II of Figure 6f and major structures for G_3 in Figure 5d), the broader distribution region also contains low probability conformations, in which the glyme is open and partially coordinated to Li^+ . The conformational distribution of G_4 contrasts with the other two glymes. For example, the distribution of r_t vs a_3 (Figure 6c) presents a peak at $r_t = 4.2$ Å with a value of 84° to 104° for a_3 , indicative of the glyme molecule being fully wrapped around Li^+ . These two distributions correspond to the same structure III of Figure 6f, where the occurrence of one angle is linked to a conformation in which the other angle is fixed at the other value. The previously described conformation is the one adopted by G_4 in the most predominant solvation shells (Figure 5d). However, the most common value of a_3 is $\sim 104^\circ$, as can be observed in the joint distribution (see Figure S5 of the SI) and from the marginal probability for a_3 . Marginal probability tails originating from those conformations can be observed in the conditional profiles for r_t and a_3 , which arise from the partial coordination of G_4 molecules with the lithium center. Low-probability regions are showcased with the G_4 conformation IV (Figure 6f), where the glyme is open and partially coordinated with Li^+ in one of its ends. The G_4 conformation from the point of view of the central angle (a_4) showcases a single important peak that occurs at $r_t = 4.2$ Å and $a_4 = 100^\circ$, which is the most common glyme solvation of Li^+ for G_4 (conformation III of Figure 6f). Furthermore, the observed interaction between Li^+ and G_4 is consistent with a coordination number of ~ 5 , as observed in Figure 3. The joint probability distribution of a_3 vs a_4 for G_4 can be found in

Figure S5 of the SI. Overall, G_3 and G_4 have distinct probabilities (amplitudes) for acquiring specific conformations (tail-to-tail distances and angles combinations) as indicated by the peaks and shoulders in the marginal distributions. The most probable angle conformations of G_4 closely matches those of G_3 , but with the presence of a richer conformational landscape in G_4 due to its longer chain and greater number of oxygen atoms, that confers more flexibility and oxygen sites to coordinate with Li^+ . Overall, the marginal distribution for G_3 aligns well with prior reports, but those for G_4 show significant differences.¹¹ As previously stated, the discrepancies could arise from the use of force fields that might overestimate some of the interactions.^{11,44,47} A summary of the relevant aspects found in this analysis can be found in Table 1.

Dynamical Properties. The dynamical properties of SILs were first investigated using the translational diffusion properties of the system. To this end, the mean squared displacements (MSDs) for the three SILs and coefficients of self-diffusion were calculated for the various species (Table 2 and Figure S6 of the SI). A comparison with experimental values (Table 2), reported at a SIL concentration and temperature of 303.15 K,²⁰ reveals the well-established phenomenon in which nonpolarizable force field used in this study produces diffusion constants that are an order of magnitude slower than the experiment.^{13,44} However, the ratios between diffusivities align well with the experimental findings and the largest deviations are observed for the G_2 SIL. Moreover, in the case of G_2 the simulations also observed the same deviation in the ratio of diffusivities between glyme and Li^+ .

To further assess the dynamical behavior of the SILs, the diffusive behavior was further analyzed by assuming that MSD follows a time dependence of the form t^β , where t stands for the lag time.⁴⁸ In this case, a value of $\beta = 1$ indicates a diffusion governed by Brownian motion, whereas $\beta < 1$ suggests subdiffusive dynamics. The simulation results demonstrate subdiffusive behavior for all the systems considered (Figure S6 of the SI) throughout the entire production phase. Values reported in Table 2 are referred to as “diffusion coefficients” or “self-diffusivities” in this work, even though when they were obtained from simulations in the subdiffusive regime. However, a clear trend is observed for the G_3 and G_4 systems,

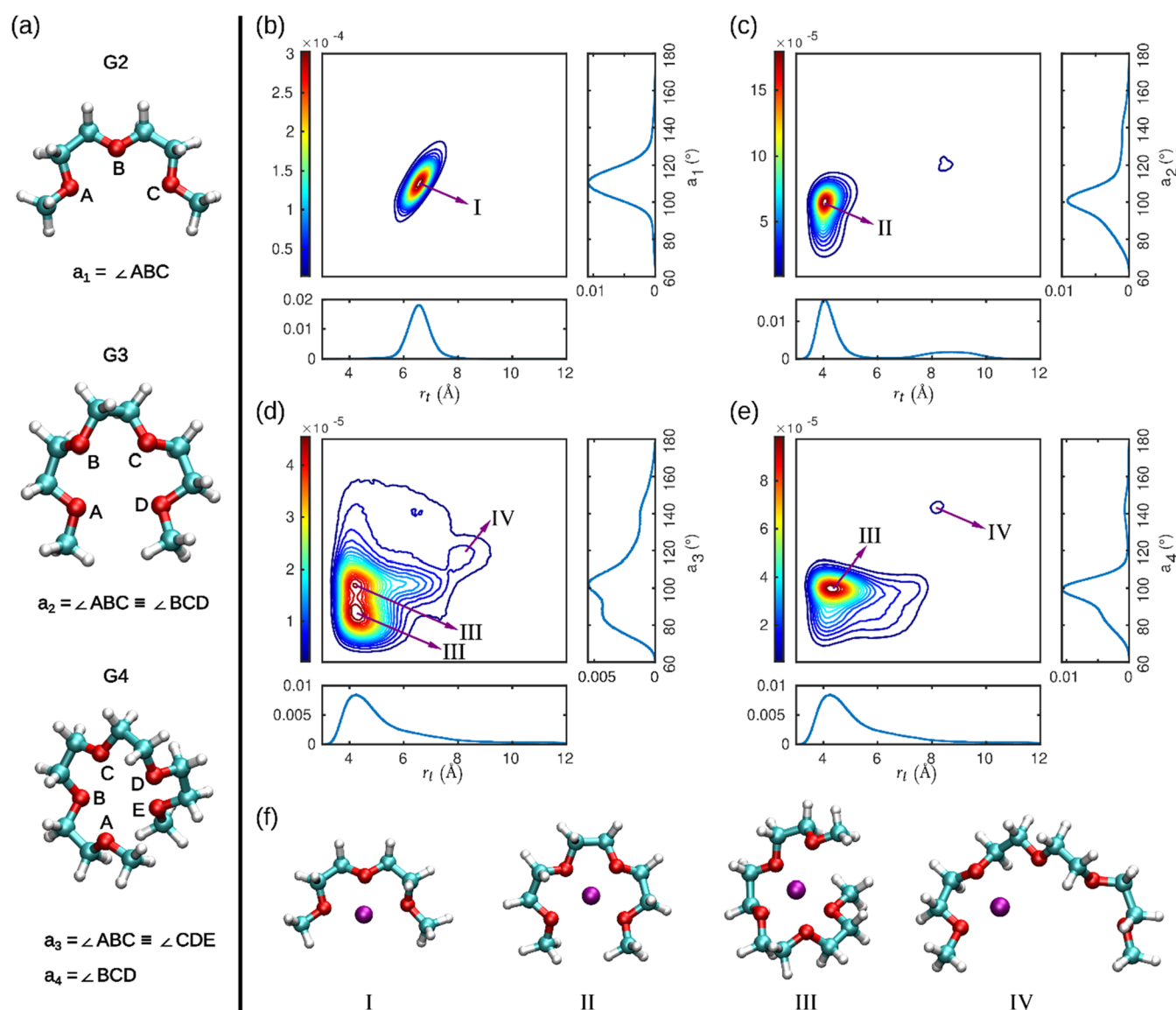


Figure 6. (a) Schematic definition of angles between oxygen atoms in glyme molecules. (b–e) Joint probability distributions of tail-to-tail distances and oxygen angles for (b) diglyme, (c) triglyme, and (d, e) tetraglyme in the various mixtures. (f) Representative glyme conformations corresponding to panels (b–e). TFSI[−] are omitted for clarity.

Table 1. Comparison between the Conformational Distributions Found for the Different SIL Systems

Feature	G2	G3	G4
Angle/s	a_1	a_2	a_3 and a_4
Angle distribution	Peak at $a_1 = 110^\circ$	Sharp peak at $a_2 = 110^\circ$; broader probability tail	Peaks at $a_3 = 84$ and 104° (main peak); $a_4 = 100^\circ$
Tail-to-tail (r_t) distribution	Sharp peak at $r_t = 6.5$ Å	Main peak at $r_t = 4.0$ Å; low probability region from 7–11 Å	Peak at $r_t = 4.2$ Å, with a long probability tail
General distribution characteristics	Single, sharp peak (a single, well-defined conformation)	One sharp main peak plus a broad low-probability region (indicates multiple conformers)	Multiple peaks and shoulders; richer conformational landscape
Glyme coordination to lithium	Fully coordinated via all 3 oxygen atoms (structure I)	Dominant structure is fully coordinated (structure II); negligible partially coordinated (open) structures present	Fully wrapped conformation around Li ⁺ (structure III); also minor open, partially coordinated structures (structure IV)

with their behavior approaching the $\beta = 1$ value at long lag times. The $[\text{Li}(\text{G2})_{4/3}][\text{TFSI}]$ mixture exhibits the slowest convergence with the value of β near 0.6 even after 400 ns (Figure S6 of the SI). Hence, it appears that longer simulations may be required for the G2 and G3 SILs to obtain converged coefficients of self-diffusion of these systems.⁴⁹ From the current data, it is observed that the value of the self-diffusion

coefficients depends on the degree of anion participation in the Li⁺ solvation shell, since their trend (i.e., $\text{G2} < \text{G3} < \text{G4}$) matches that seen for the amount of TFSI anions found in the Li⁺ solvation shells (Figure 5). This suggests that larger glymes, such as G4, screen more efficiently the Coulombic interaction between ions of opposite charge, enabling both the solvated cation and the anion to move more freely. In other words,

Table 2. Densities and Self-Diffusion Coefficients (Obtained in the Sub-Diffusive Regime; See Text) for the Three Species Present in the Glyme Mixtures Analyzed in This Study, along with Diffusion Coefficient Ratios Derived from Molecular Dynamics Simulations and Compared with Experimental Data^a

SIL	ρ (g/cm ³)		D ($\times 10^{-7}$ cm ² /s)								$D_{\text{glyme}}/D_{\text{Li}}$		$D_{\text{TFSI}}/D_{\text{Li}}$	
			D_{glyme}		D_{Li}		D_{TFSI}							
	MD	Exp	MD	Exp	MD	Exp	MD	Exp	MD	Exp	MD	Exp		
G2	1.52	1.40	0.017	0.48	0.015	0.47	0.018	0.41	1.1	1.02	1.2	0.87		
G3	1.55	1.42	0.035	0.77	0.035	0.77	0.038	0.54	1.0	1.0	1.1	0.70		
G4	1.50	1.40	0.16	1.26	0.15	1.26	0.18	1.22	1.0	1.0	1.2	0.97		

^aAll experimental values are from ref 20.

these simulation results show that the self-diffusion coefficients of Li⁺ are defined by the formation of ion pairs, rather than by their hydrodynamic radii.

The rotational dynamics of the anion and glyme molecules were also investigated (see the [Computational Methodology](#) section). Rotational relaxation times for TFSI[−] and the different glymes are summarized in [Table 3](#), while the

Table 3. Integrated Rotational Relaxation Times (in ns) for the TFSI[−] and Glyme Species for the Different SILs

SIL	$\langle\tau(\text{TFSI}^-)\rangle$	$\langle\tau(\text{glyme})\rangle$
G2	41.7	20.9
G3	17.6	10.9
G4	4.3	2.7

corresponding orientational time correlation function fits (eq 2) are shown in the Supporting Information ([Table S4](#) and [Figure S7](#)). From the rotational times, it is observed that glyme molecules have shorter rotational relaxation times than TFSI[−], but their trend does not follow their size: $\tau_{\text{G2}} > \tau_{\text{G3}} > \tau_{\text{G4}}$. The longest relaxation times are found for both components (TFSI[−] and G2) in the G2 SIL, suggesting that the solvation structures in the G2 system have the longest lifetimes. The G3 system shows intermediate relaxation times, approximately half of those seen for G2, while the G4 system exhibits the fastest rotational dynamics, with relaxation times approximately an order of magnitude shorter than those of G2. The behavior of the rotational motions appears to be strongly correlated to the degree in which the anion is involved in the solvation shell of Li⁺. For example, the major solvation structures of G2 primarily feature one or two glymes ([Figure 5](#)), where the anions bind tightly to Li⁺, as indicated by the single peak in the conformational map ([Figure 6b](#)). Accordingly, G2 has the lowest fraction of free anions among the three glyme systems (17.9% of the total, [Table S3](#)), which is reflected in the slowest translational and rotational dynamics of both anion and glyme ([Tables 2](#) and [3](#)). In contrast, the G3 system shows slightly higher anion mobility and a marginally increased percentage of free anions (18.4%), attributed to a weaker Li⁺-anion interaction due to a more effective chelation of Li⁺ by a single G3 molecule. The effect becomes more pronounced in the G4 system, where a significant fraction of anions (46.3%) is not in the Li⁺ solvation shell, which is a direct consequence of a more complete chelation of Li⁺ by the longer glyme chain of G4. The reduced anion coordination in the G4 SIL is reflected in their increased mobility, as seen in the self-diffusivity and rotational relaxation times of both TFSI and G4. Finally, the rotational relaxation times for the anion and glymes ([Table 3](#)) exhibit inverse relationships with their respective self-

diffusivities ([Table 2](#)) suggesting the presence of a microscopic mechanism connecting the two.

Thermal and Electrochemical Stability. Yoshida et al. carried out thermal stability studies on Li-SILs using thermal gravimetric (TG) analysis. They concluded that the initial weight loss in these SILs in the range of 200 to 400 °C was due to the evaporation of solvent.⁵⁰ MD studies of water evaporation have shown that the free energy change for moving a single water molecule across the liquid–vapor interface reflects the cohesiveness of the liquid relative to its vapor state; in aqueous salt solutions, this free energy increases with salt concentration.^{51,52} By analogy, the free energy required to bring a glyme in the bulk SIL to the vapor is a suitable qualitative measure of the thermal stability of the SILs. In the present work, the thermal stability of the SILs was therefore assessed by computing the free energy change associated with transferring a glyme molecule from the bulk solution to the gas phase (see [Computational Methodology](#) section and [Figure 2](#)). These free energy profiles are shown in [Figure 7](#), with the zero-energy reference set at the center of the

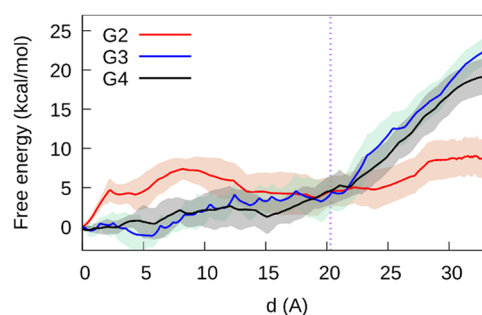


Figure 7. Free energy profiles for the different systems, with shaded regions indicating the error bars for the free energy values. The interface, shown as a purple dashed line, corresponds to the average region where the density drops to half of its bulk value.

bulk electrolyte. The profile for G2 exhibits an initial sharp energy increase, corresponding to the breaking of the initial Li⁺ solvation structure involving two diglyme molecules. As the glyme approaches the interface, additional glyme recombination events further shape its energy profile. In contrast, the G3 and G4 profiles display a gradual rise in energy from the bulk toward the vacuum region, where a slight difference between them becomes apparent. Notably, all three profiles converge near the electrolyte interface, located around 20.3 Å from the bulk center. The energy barrier between the fully solvated glyme to free in gas phase is used as an indicator of the thermal stability of the glyme molecules in the SILs. The energy difference between the final and initial states for the

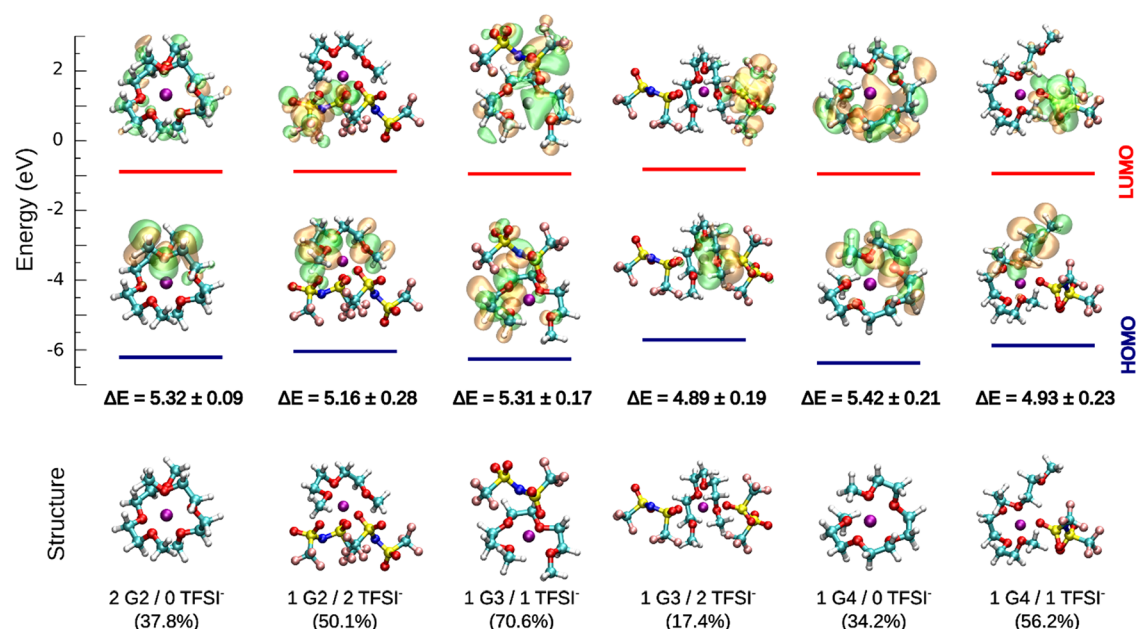


Figure 8. HOMO–LUMO energy gaps ΔE for the major structures of the glyme-based electrolytes studied in this work. Abundance percentages of the major structures inside the respective electrolytes are also presented. The isosurface values for the orbitals were set to ± 0.02 au.

[Li(G3)₁][TFSI] SIL is 24 ± 2 kcal/mol, 20 ± 2 kcal/mol for [Li(G4)₁][TFSI] case and the smallest energy difference, 9 ± 3 kcal/mol, correspond to the [Li(G2)_{4/3}][TFSI] SIL, indicating that the latter has the lowest thermal stability for all the studied SILs. These thermodynamics stabilities agree well with the TG analysis on the G2 SIL²⁰ and the TG experiments performed for the G3 and G4 SILs.^{50,53} In summary, the MD simulations predict that the G2 SIL is the least thermally stable sample, which is in agreement with the experiments. In contrast, the simulations predict an almost similar free energy barrier for removing a solvent molecule to the gas phase, within the error bars of the calculation, between the bound and free G3 and G4 cases, which is in contrast with the slightly higher stability for G4 compared to G3 observed experimentally.^{50,53}

It is important to note that the predicted thermal stabilities presented in Figure 7 are rather qualitative due to the subdiffusive behavior observed for G2 and G3, which limits the sampling of phase space in all-atom simulations (even with the use of enhanced sampling methods such as umbrella sampling). Nevertheless, these are sufficient to capture the experimental trends in the thermal stability of these SILs, providing us with a molecular framework to explore the molecular origin of the thermal stability. For example, the lower thermal stability of the [Li(G2)_{4/3}][TFSI] electrolyte can be attributed to the weaker coordination of the G2 molecule to Li⁺ through its three oxygen atoms, compared to the stronger coordination in G3 and G4, where four and five glyme oxygen atoms anchor to Li⁺, respectively. This confirms a previous hypothesis in which the instability of the complex [Li(G2)₂]⁺ cation was attributed to the easy evaporation of G2 molecules in the [Li(G2)₂][TFSI] system at temperatures higher than the complex melting point.¹⁹ On the other hand, the formation of structures around Li⁺ involving two G2 molecules in [Li(G2)_{4/3}][TFSI] screens the interaction between Li⁺ and TFSI⁻, leading to a non-negligible fraction of TFSI⁻ ions that do not participate in the first solvation shell, and to other structures involving one G2 and two or more

TFSI⁻ (Figure 5). Hence, the simulation results indicate that the TFSI⁻ anion also plays an important role in the behavioral thermal stability of the G2 system. In contrast, the preferred first solvation shell structure for Li⁺ in G3 consists of one glyme molecule and one TFSI⁻ anion, resulting in a net CN of ~ 5 , with the glyme contributing the majority of coordinating oxygen atoms (Figures 3 and 5). This trend is even more pronounced in the case of G4, where the TFSI⁻ provides 0.8 of the total 5.4 coordination sites. The longer chain of G4 allows it to fully coordinate the Li⁺ ion, minimizing the involvement of the anion. Overall, thermal stability appears to be directly correlated with both the degree of Li⁺ chelation per glyme and the participation of the anion in the solvation shell. A subtle balance between these two factors seems to account for the slightly higher stability of [Li(G3)₁][TFSI] compared to [Li(G4)₁][TFSI] in the simulations.

Electrochemical stability is another key characteristic of the investigated SILs.^{20,53,54} It has been reported that, at SIL concentrations, the oxidative stability of glyme molecules is significantly enhanced when they form complex cations, such as [Li(glyme)₁]⁺. The coordination lowers the HOMO energy level of the oligoether solvents, thereby suppressing oxidative decomposition.^{20,53} HOMO and LUMO energies of the dominant solvation structures were calculated for the various glyme-based systems, as the energy gap between these frontier molecular orbitals is closely associated with the redox potential and offers valuable insight into the electrochemical stability of the electrolyte.^{42,55–57} The corresponding energy for the HOMO and LUMO levels, along with the spatial distribution of the orbitals, are shown in Figure 8.

The computations reveal that the HOMO is localized on the solvent molecules, while the LUMO is primarily located on the anions when present, a trend also observed in other lithium electrolytes.⁵⁵ LUMO energies are all similar and close to -1 eV, whereas slightly larger differences can be observed for the HOMO levels (Figure 8). The results presented in Figure 8 show that the major solvation structures, accounting for $\sim 90\%$ of the total species, exhibit similar HOMO–LUMO gap

energies of around 5.0 eV. The structures with the smallest HOMO–LUMO gaps establish the lower bound of the electrochemical stability window for each system, and interestingly, the structures exhibiting the lowest energy gaps correspond to those containing anions. Therefore, the G2 electrolyte seems to be slightly more stable than G3 and G4, with the stability of the latter two being nearly identical. Overall, the calculated energy gaps agree well with experimental results from linear sweep voltammetry, where oxidative stability potentials are consistently observed around 5 V vs Li⁺/Li in these electrolytes.^{20,53,54}

Despite significant advancements, the speciation and dynamics of lithium solvation shells in SILs, as well as the mechanisms underlying ionic conduction, are still not fully understood from a fundamental perspective. The findings of this study provide a detailed exploration of key properties such as ion diffusion, coordination environments, and molecular structure of Li-SILs. Previous work has shown that in sodium based glyme electrolytes, the solvation structures near electrodes were similar to the bulk electrolytes since chelation rather than dielectric constant determines the electrolyte behavior.⁵⁸ This chelation effect also applies to the Li-SIL system. Thus, investigations on electrolyte decomposition pathways can now focus on the dominant solvation structures identified in this work rather than complex computations involving electrolyte-electrode interfaces. Finally, these Li-SILs are a relatively new class of electrolytes, which despite similarities to ionic liquids have a very different intermolecular interaction (chelation versus electrostatic effects) and thus merit closer investigations.

SUMMARY

Solvate ionic liquids (SILs) based on glymes and lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) were simulated using a nonpolarizable force field specifically tailored for these systems. Solvation shells and coordination fractions for both glymes and anions relative to Li⁺ were determined, revealing almost no free glymes in the electrolytes. In the [Li(G2)_{4/3}]-[TFSI] system, structures featuring one and two G2 glyme molecules coordinated with lithium were found to coexist in similar amounts. In contrast, G3 and G4 glymes coordinate Li⁺ predominantly with one solvent molecule, which is consistent with experiments. The results for [Li(G2)_{4/3}][TFSI] and [Li(G3)]₁[TFSI] also reveal similar Li–O coordination numbers for both solvent and anion, despite notable differences in the solvent structure.

Solvent conformational analysis, based on tail-to-tail distances and angles between oxygen atoms, shows a well-defined geometry for G2 at SIL concentration. In comparison, G3 presents a less rigid conformation. The G4 system displays an even more complex conformational landscape, reflecting the increased flexibility afforded by its longer chain. Translational diffusion, such as self-diffusivities, are in good agreement with experimental data, while rotational relaxation times for both glymes and TFSI follow a trend similar to their corresponding diffusion coefficients. In addition, subdiffusive behavior is observed for both G2 and G3 SILs, but not in G4.

The thermal stability of the studied SILs from simulations, assessed via the free energy profile of moving a glyme from the bulk electrolyte to the air interface, shows that the G3 and G4 mixtures are more stable than the G2 system, in agreement with thermogravimetric analysis results. Furthermore, the computed electrochemical stability (evaluated via HOMO–

LUMO gaps) is found to follow the trends consistent with cyclic voltammetry experiments. In the latter case, simulations show that the process is governed by the presence of anions in the solvation shell of Li⁺. Overall, the findings of the study underscore the importance of the anion in influencing the dynamical behavior as well as the thermal and electrochemical stability of these SILs and SIL-like electrolytes. The methodology used here provides a molecular framework for predicting the dynamical and macroscopic properties of SIL electrolytes.

ASSOCIATED CONTENT

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.5c04265>.

Details on the number of glyme molecules and ion pairs used in the different simulations, a comparison of RDFs, percentage of Li⁺ coordinated to TFSI[−], plots of mean square displacements and corresponding β values, simulated X-ray partial structure factors, orientational correlation functions, as well as position distributions and free energy profiles obtained through the umbrella sampling method (PDF).

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Tamura, T.; Hachida, T.; Yoshida, K.; Tachikawa, N.; Dokko, K.; Watanabe, M. New glyme–cyclic imide lithium salt complexes as thermally stable electrolytes for lithium batteries. *J. Power Sources* 2010, 195 (18), 6095–6100.

- (2) Ueno, K.; Tatara, R.; Tsuzuki, S.; Saito, S.; Doi, H.; Yoshida, K.; Mandai, T.; Matsugami, M.; Umebayashi, Y.; Dokko, K.; Watanabe, M. Li⁺ solvation in glyme–Li salt solvate ionic liquids. *Phys. Chem. Chem. Phys.* **2015**, *17* (12), 8248–8257.
- (3) Yoshida, K.; Tsuchiya, M.; Tachikawa, N.; Dokko, K.; Watanabe, M. Correlation between battery performance and lithium ion diffusion in glyme–lithium bis (trifluoromethanesulfonyl) amide equimolar complexes. *J. Electrochem. Soc.* **2012**, *159* (7), No. A1005.
- (4) Henderson, W. A. Glyme–lithium salt phase behavior. *J. Phys. Chem. B* **2006**, *110* (26), 13177–13183.
- (5) Aguilera, L.; Xiong, S.; Scheers, J.; Matic, A. A structural study of LiTFSI–tetraglyme mixtures: From diluted solutions to solvated ionic liquids. *J. Mol. Liq.* **2015**, *210*, 238–242.
- (6) Martinelli, A.; Matic, A.; Jacobsson, P.; Borjesson, L.; Fernicola, A.; Scrosati, B. Phase behavior and ionic conductivity in lithium bis (trifluoromethanesulfonyl) imide-doped ionic liquids of the pyrrolidinium cation and bis (trifluoromethanesulfonyl) imide anion. *J. Phys. Chem. B* **2009**, *113* (32), 11247–11251.
- (7) Zhang, C.; Gu, S.; Zhang, D.; Ma, J.; Zheng, H.; Zheng, M.; Lv, R.; Yu, K.; Wu, J.; Wang, X.; et al. Nonflammable, localized high-concentration electrolyte towards a high-safety lithium metal battery. *Energy Storage Mater.* **2022**, *52*, 355–364.
- (8) Jiang, G.; Li, F.; Wang, H.; Wu, M.; Qi, S.; Liu, X.; Yang, S.; Ma, J. Perspective on high-concentration electrolytes for lithium metal batteries. *Small Struct.* **2021**, *2* (5), No. 2000122.
- (9) Li, M.; Wang, C.; Chen, Z.; Xu, K.; Lu, J. New concepts in electrolytes. *Chem. Rev.* **2020**, *120* (14), 6783–6819.
- (10) Carrillo-Bohórquez, O.; Kuroda, D. G.; Kumar, R. A transferable classical force field to describe glyme based lithium solvate ionic liquids. *J. Chem. Phys.* **2024**, *161* (5), No. 054504, DOI: 10.1063/5.0214769.
- (11) Shinoda, W.; Hatanaka, Y.; Hirakawa, M.; Okazaki, S.; Tsuzuki, S.; Ueno, K.; Watanabe, M. Molecular dynamics study of thermodynamic stability and dynamics of [Li (glyme)]⁺ complex in lithium-glyme solvate ionic liquids. *J. Chem. Phys.* **2018**, *148* (19), No. 193809, DOI: 10.1063/1.5009814.
- (12) Carrillo-Bohórquez, O.; Kumar, R.; Kuroda, D. G. The Challenge of Characterizing High-Concentration Electrolytes at the Molecular Level: A Perspective. *J. Phys. Chem. C* **2024**, *128* (44), 18695–18710.
- (13) Dong, D.; Bedrov, D. Charge transport in [Li (tetraglyme)][bis (trifluoromethane) sulfonimide] solvate ionic liquids: insight from molecular dynamics simulations. *J. Phys. Chem. B* **2018**, *122* (43), 9994–10004.
- (14) Li, K.; Don, V. S.; Gupta, C. S.; David, R.; Kumar, R. Effect of anion identity on ion association and dynamics of sodium ions in non-aqueous glyme based electrolytes—OTf vs TFSI. *J. Chem. Phys.* **2021**, *154* (18), No. 184505, DOI: 10.1063/5.0046073.
- (15) Tsuzuki, S.; Shinoda, W.; Matsugami, M.; Umebayashi, Y.; Ueno, K.; Mandai, T.; Seki, S.; Dokko, K.; Watanabe, M. Structures of [Li (glyme)]⁺ complexes and their interactions with anions in equimolar mixtures of glymes and Li [TFSI]: analysis by molecular dynamics simulations. *Phys. Chem. Chem. Phys.* **2015**, *17* (1), 126–129.
- (16) Shimizu, K.; Freitas, A. A.; Atkin, R.; Warr, G. G.; FitzGerald, P. A.; Doi, H.; Saito, S.; Ueno, K.; Umebayashi, Y.; Watanabe, M.; Lopes, J. N. C. Structural and aggregate analyses of (Li salt+ glyme) mixtures: the complex nature of solvate ionic liquids. *Phys. Chem. Chem. Phys.* **2015**, *17* (34), 22321–22335.
- (17) Murphy, T.; Callear, S. K.; Yepuri, N.; Shimizu, K.; Watanabe, M.; Lopes, J. N. C.; Darwish, T.; Warr, G. G.; Atkin, R. Bulk nanostructure of the prototypical ‘good’ and ‘poor’ solvate ionic liquids [Li (G4)][TFSI] and [Li (G4)][NO 3]. *Phys. Chem. Chem. Phys.* **2016**, *18* (26), 17224–17236.
- (18) Philipp, J. K.; Fumino, K.; Appellagen, A.; Paschek, D.; Ludwig, R. The Competition Between Cation-Anion and Cation-Triglyme Interaction in Solvate Ionic Liquids Probed by Far Infrared Spectroscopy and Molecular Dynamics Simulations. *ChemPhysChem* **2025**, *26*, No. e202400991.
- (19) Mandai, T.; Yoshida, K.; Ueno, K.; Dokko, K.; Watanabe, M. Criteria for solvate ionic liquids. *Phys. Chem. Chem. Phys.* **2014**, *16* (19), 8761–8772.
- (20) Zhang, C.; Ueno, K.; Yamazaki, A.; Yoshida, K.; Moon, H.; Mandai, T.; Umebayashi, Y.; Dokko, K.; Watanabe, M. Chelate effects in glyme/lithium bis (trifluoromethanesulfonyl) amide solvate ionic liquids. I. Stability of solvate cations and correlation with electrolyte properties. *J. Phys. Chem. B* **2014**, *118* (19), 5144–5153.
- (21) Choquette, Y.; Brisard, G.; Parent, M.; Brouillette, D.; Perron, G.; Desnoyers, J. E.; Armand, M.; Gravel, D.; Slougui, N. Sulfamides and glymes as aprotic solvents for lithium batteries. *J. Electrochem. Soc.* **1998**, *145* (10), No. 3500.
- (22) Qian, K.; Seifert, S.; Winans, R. E.; Li, T. Understanding solvation behavior of the saturated electrolytes with small/wide-angle X-ray scattering and Raman spectroscopy. *Energy Fuels* **2021**, *35* (23), 19849–19855.
- (23) Arai, N.; Watanabe, H.; Yamaguchi, T.; Seki, S.; Ueno, K.; Dokko, K.; Watanabe, M.; Kameda, Y.; Buchner, R.; Umebayashi, Y. Dynamic chelate effect on the Li⁺-ion conduction in solvate ionic liquids. *J. Phys. Chem. C* **2019**, *123* (50), 30228–30233.
- (24) Thompson, A. P.; Aktulga, H. M.; Berger, R.; Bolintineanu, D. S.; Brown, W. M.; Crozier, P. S.; In’t Veld, P. J.; Kohlmeyer, A.; Moore, S. G.; Nguyen, T. D.; et al. LAMMPS—a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. *Comput. Phys. Commun.* **2022**, *271*, No. 108171.
- (25) Martínez, L.; Andrade, R.; Birgin, E. G.; Martínez, J. M. PACKMOL: A package for building initial configurations for molecular dynamics simulations. *J. Comput. Chem.* **2009**, *30* (13), 2157–2164.
- (26) Humphrey, W.; Dalke, A.; Schulten, K. VMD: visual molecular dynamics. *J. Mol. Graphics* **1996**, *14* (1), 33–38.
- (27) Brehm, M.; Thomas, M.; Gehrke, S.; Kirchner, B. TRAVIS—A free analyzer for trajectories from molecular simulation. *J. Chem. Phys.* **2020**, *152* (16), No. 164105, DOI: 10.1063/5.0005078.
- (28) Brehm, M.; Kirchner, B. TRAVIS—A Free Analyzer and Visualizer for Monte Carlo and Molecular Dynamics Trajectories; ACS Publications, 2011.
- (29) Tao, T. Time-dependent fluorescence depolarization and Brownian rotational diffusion coefficients of macromolecules. *Biopolymers* **1969**, *8* (5), 609–632.
- (30) Li, M.; Owruksky, J.; Sarisky, M.; Culver, J.; Yodh, A.; Hochstrasser, R. Vibrational and rotational relaxation times of solvated molecular ions. *J. Chem. Phys.* **1993**, *98* (7), 5499–5507.
- (31) Wong, V.; Case, D. A. Evaluating rotational diffusion from protein MD simulations. *J. Phys. Chem. B* **2008**, *112* (19), 6013–6024.
- (32) Annapureddy, H. V. R.; Kashyap, H. K.; De Biase, P. M.; Margulis, C. J. What is the origin of the prepeak in the X-ray scattering of imidazolium-based room-temperature ionic liquids? *J. Phys. Chem. B* **2010**, *114* (50), 16838–16846.
- (33) Fiorin, G.; Klein, M. L.; Hénin, J. Using collective variables to drive molecular dynamics simulations. *Mol. Phys.* **2013**, *111* (22–23), 3345–3362.
- (34) Kumar, S.; Rosenberg, J. M.; Bouzida, D.; Swendsen, R. H.; Kollman, P. A. The weighted histogram analysis method for free-energy calculations on biomolecules. I. The method. *J. Comput. Chem.* **1992**, *13* (8), 1011–1021.
- (35) Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Scalmani, G.; Barone, V.; Petersson, G. A.; Nakatsuji, H.; et al. *Gaussian 16*, Rev. C.01; Gaussian Inc.: Wallingford, CT, 2016.
- (36) Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized gradient approximation made simple. *Phys. Rev. Lett.* **1996**, *77* (18), No. 3865.
- (37) Frisch, M. J.; Pople, J. A.; Binkley, J. S. Self-consistent molecular orbital methods 25. Supplementary functions for Gaussian basis sets. *J. Chem. Phys.* **1984**, *80* (7), 3265–3269.
- (38) Borodin, O.; Olguin, M.; Ganesh, P.; Kent, P. R.; Allen, J. L.; Henderson, W. A. Competitive lithium solvation of linear and cyclic

carbonates from quantum chemistry. *Phys. Chem. Chem. Phys.* **2016**, *18* (1), 164–175.

(39) Barone, V.; Cossi, M. Quantum calculation of molecular energies and energy gradients in solution by a conductor solvent model. *J. Phys. Chem. A* **1998**, *102* (11), 1995–2001.

(40) Cossi, M.; Rega, N.; Scalmani, G.; Barone, V. Energies, structures, and electronic properties of molecules in solution with the C-PCM solvation model. *J. Comput. Chem.* **2003**, *24* (6), 669–681.

(41) Huang, M.-M.; Jiang, Y.; Sasisanker, P.; Driver, G. W.; Weingärtner, H. Static relative dielectric permittivities of ionic liquids at 25 °C. *J. Chem. Eng. Data* **2011**, *56* (4), 1494–1499.

(42) Méndez-Hernández, D. D.; Gillmore, J. G.; Montano, L. A.; Gust, D.; Moore, T. A.; Moore, A. L.; Mujica, V. Building and testing correlations for the estimation of one-electron reduction potentials of a diverse set of organic molecules. *J. Phys. Org. Chem.* **2015**, *28* (5), 320–328.

(43) Henderson, W. A.; McKenna, F.; Khan, M. A.; Brooks, N. R.; Young, V. G., Jr; Frech, R. Glyme–lithium bis (trifluoromethanesulfonyl) imide and glyme–lithium bis (perfluoroethanesulfonyl) imide phase behavior and solvate structures. *Chem. Mater.* **2005**, *17* (9), 2284–2289.

(44) Harte, T.; Dharmasiri, B.; Dobhal, G. S.; Walsh, T. R.; Henderson, L. C. Accelerated lithium-ion diffusion via a ligand ‘hopping’ mechanism in lithium enriched solvate ionic liquids. *Phys. Chem. Chem. Phys.* **2023**, *25* (43), 29614–29623.

(45) Qian, K.; Winans, R. E.; Li, T. Insights into the nanostructure, solvation, and dynamics of liquid electrolytes through small-angle x-ray scattering. *Adv. Energy Mater.* **2021**, *11* (4), No. 2002821.

(46) Saito, S.; Watanabe, H.; Ueno, K.; Mandai, T.; Seki, S.; Tsuzuki, S.; Kameda, Y.; Dokko, K.; Watanabe, M.; Umebayashi, Y. Li⁺ local structure in hydrofluoroether diluted Li-glyme solvate ionic liquid. *J. Phys. Chem. B* **2016**, *120* (13), 3378–3387.

(47) Vigil, D. L.; Stevens, M. J.; Frischknecht, A. L. Accurate Calculation of Solvation Properties of Lithium Ions in Nonaqueous Solutions. *J. Phys. Chem. B* **2023**, *127* (37), 8002–8008.

(48) Del Pópolo, M. G.; Voth, G. A. On the structure and dynamics of ionic liquids. *J. Phys. Chem. B* **2004**, *108* (5), 1744–1752.

(49) Maginn, E. J. Atomistic simulation of the thermodynamic and transport properties of ionic liquids. *Acc. Chem. Res.* **2007**, *40* (11), 1200–1207.

(50) Yoshida, K.; Tsuchiya, M.; Tachikawa, N.; Dokko, K.; Watanabe, M. Change from glyme solutions to quasi-ionic liquids for binary mixtures consisting of lithium bis (trifluoromethanesulfonyl) amide and glymes. *J. Phys. Chem. C* **2011**, *115* (37), 18384–18394.

(51) Varilly, P.; Chandler, D. Water evaporation: A transition path sampling study. *J. Phys. Chem. B* **2013**, *117* (5), 1419–1428.

(52) Pandey, P. K.; Chandra, A. Mechanism, kinetics, and potential of mean force of evaporation of water from aqueous sodium chloride solutions of varying concentrations. *J. Phys. Chem. B* **2023**, *127* (20), 4602–4612.

(53) Yoshida, K.; Nakamura, M.; Kazue, Y.; Tachikawa, N.; Tsuzuki, S.; Seki, S.; Dokko, K.; Watanabe, M. Oxidative-stability enhancement and charge transport mechanism in glyme–lithium salt equimolar complexes. *J. Am. Chem. Soc.* **2011**, *133* (33), 13121–13129.

(54) Di Lecce, D.; Marangon, V.; Jung, H.-G.; Tominaga, Y.; Greenbaum, S.; Hassoun, J. Glyme-based electrolytes: suitable solutions for next-generation lithium batteries. *Green Chem.* **2022**, *24* (3), 1021–1048.

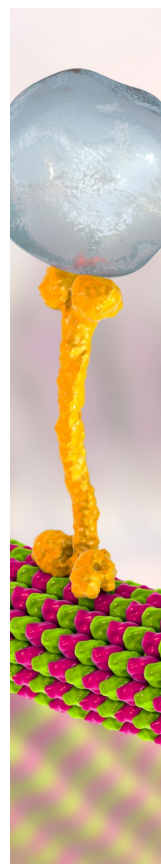
(55) Pham, T. D.; Faheem, A. B.; Kim, J.; Oh, H. M.; Lee, K. K. Practical High-Voltage Lithium Metal Batteries Enabled by Tuning the Solvation Structure in Weakly Solvating Electrolyte. *Small* **2022**, *18* (14), No. 2107492.

(56) Peljo, P.; Girault, H. H. Electrochemical potential window of battery electrolytes: the HOMO–LUMO misconception. *Energy Environ. Sci.* **2018**, *11* (9), 2306–2309.

(57) Gauthier, M.; Carney, T. J.; Grimaud, A.; Giordano, L.; Pour, N.; Chang, H.-H.; Fenning, D. P.; Lux, S. F.; Paschos, O.; Bauer, C.; et al. Electrode–electrolyte interface in Li-ion batteries: current

understanding and new insights. *J. Phys. Chem. Lett.* **2015**, *6* (22), 4653–4672.

(58) Senadheera, D. I.; Carrillo-Bohorquez, O.; Nachaki, E. O.; Jorn, R.; Kuroda, D. G.; Kumar, R. Probing the Electrode–Electrolyte Interface of Sodium/Glyme-Based Battery Electrolytes. *J. Phys. Chem. C* **2024**, *128* (14), 5798–5808.



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